RESEARCH STATEMENT

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I am a macroeconomist working at the intersection of business cycles and information economics. Broadly, my research seeks to understand how expectations are formed and how the resulting beliefs shape economic outcomes. This note provides an overview of my work in three parts. *Rational Inattention in Macroeconomics* covers how decision-makers allocate costly attention across the economy, how that allocation shapes their actions and aggregate dynamics, and how models with such agents can be solved. *Learning in Dynamic Models* turns to a newer line of work on how agents learn about their environment from the data they observe, and what that learning implies for the transmission of shocks in general equilibrium. *Other Research* covers projects that fall into neither category.

Rational Inattention in Macroeconomics

Rational Inattention (RI), introduced by Sims (1998, 2003), starts from the premise that every economic decision is first an attention problem: before acting, decision-makers must decide what to learn about the state of the world. Acquiring more information improves decisions but requires cognitive effort, which RI models as a cost and trades off against that benefit. Concretely, agents design signals: they choose what to pay attention to and how precisely. Because attention is costly, these signals are noisy, and filtering them induces inertia in their decisions (e.g., prices, consumption, investment, etc). In the following papers, I embed RI in different Dynamic Stochastic General Equilibrium (DSGE) models and study how inattention helps close the gap between their predictions and the data. I also develop a method to solve such RI-DSGE models efficiently.

Inattention in Estimated Business Cycle Models

In *"Macroeconomic Shocks & Their Propagation under Rational Inattention,"* I construct and estimate a medium-scale DSGE model in which firms and households are rationally inattentive. To the best of my knowledge, my model is the first RI-DSGE to feature both pricing and capital decisions, and to be estimated with Bayesian methods.[1] One of the main results of the paper is that, in this setup, inattention can fit the data only when paired with a real friction affecting capital accumulation. The reason is easiest to see after a monetary policy shock: inattention induces slow-adjusting prices, which cause a contraction in aggregate demand; absent any other friction, the contraction produces a large drop in firms' optimal capital stock, which draws their attention. Firms therefore track the optimal capital stock closely, so investment dynamics are peaked on impact rather than hump-shaped. General equilibrium feedback amplifies the volatility of the optimal capital stock, which only strengthens firms' incentive to track it. Thus, no degree of inattention alone can deliver inertia in both prices and investment.

[1] Maćkowiak and Wiederholt (2015) feature pricing but no capital decisions; Maćkowiak and Wiederholt (2025) feature capital but no pricing.

I estimate the model with capital adjustment costs using quasi-Bayesian methods, by matching the impulse responses to a monetary policy shock. The model fits the evidence as well as a prototypical New Keynesian benchmark and is even favored by the marginal data density. When I feed the shock processes estimated with the New Keynesian model into the RI model, the two models nearly coincide in their responses to technology and investment shocks—untargeted, and the main drivers of the business cycle—but diverge sharply on markup shocks, long criticized as reduced-form wedges. Lastly, the RI model, in which attention is endogenous, reproduces the large, impact-peaked firm-level responses to idiosyncratic shocks documented in micro data, a pattern the New Keynesian firm cannot generate.

Solving Models with Rational Inattention

RI-DSGE models have so far remained confined to a handful of papers, arguably not for lack of interest: solving them requires finding a fixed point between agents' attention choices and the equilibrium dynamics these choices induce, which requires considerable know-how and computing time. In *"Solving Dynamic Models with Rational Inattention,"* I propose a new algorithm that decouples attention from the general equilibrium. Given attention, an inner loop solves exactly for the paths of all endogenous variables in the sequence space, by mapping the solution of the attention problems into expectation matrices; an outer loop then updates attention given those paths. Existing procedures instead revise a guess of the aggregate paths jointly with attention, so general equilibrium feedbacks are resolved only across iterations and the iteration remains stable only with very small steps. On the models from the literature, my algorithm recovers the published equilibria 2 to almost 2,000 times faster, which makes likelihood-based estimation of RI-DSGE models feasible. The accompanying toolbox aims to make these models accessible beyond specialists.

Inattention & Household Heterogeneity

A natural extension is to take RI to models with household heterogeneity, in which the stakes of paying attention differ across households. Endogenous attention can then explain the systematic differences in expectations across household characteristics documented in survey data (Mitman et al. 2022).Those expectations in turn feed back into households' decisions. *"Monetary Policy with Inattentive Heterogeneous Households"* pursues this direction.[2]

I build a tractable model with inattentive heterogeneous households that matches the iMPC profile of its HANK counterpart. The estimated model matches both the aggregate response to a monetary policy shock and the accuracy of inflation expectations across the wealth distribution, and serves to study the nexus between policies—fiscal and monetary—and endogenous expectations formation.

Learning in Dynamic Models

Rational expectations impose that agents' beliefs are consistent with the equilibrium of the model: agents know the laws of motion of the economy and forecast accordingly. Models of

[2]This paper is a work in progress that supersedes a previously circulated version titled *"Rationally Inattentive Heterogeneous Households."*

learning relax this assumption (Evans and Honkapohja, 2001). Agents instead form beliefs from the data they observe, for instance by estimating forecasting models that they update as new observations arrive. The economy no longer sits at a single fixed point, but converges toward the rational expectations equilibrium as agents learn, and beliefs and outcomes feed back into each other along the way. Learning can thereby generate richer dynamics, such as additional persistence and amplification of shocks, that rational expectations models need other frictions to produce (Slobodyan and Wouters, 2012).

Sparse Learning

Models of sparse learning (Gabaix 2014, 2016), in which agents simplify their world and decisions by shrinking the weights they place on different state variables, have seen few macroeconomic applications. Yet the framework is naturally close to more complex and more established models, such as RI, in which sparsity emerges from the optimal allocation of attention, while being far cheaper to solve. In a paper tentatively titled *"Learning under Constraint: Implications for Macroeconomics"*, with Sergey Slobodyan (CERGE-EI), we show that under certain assumptions sparse learning is equivalent, or very close, to RI, so that RI can be read as a form of sparse learning. We then exploit this correspondence to combine costly attention with standard adaptive learning: agents choose which state variables to attend to while also learning about the parameters of their environment from the data. Unlike in steady-state RI, where attention is time-invariant, attention and learning then shape each other: what agents learn changes what they attend to, and what they attend to changes what they learn. We estimate DSGE models with both features.

Learning in Heterogeneous Agent Models

In heterogeneous-agent (HA) models with perfect information rational expectations, agents forecast prices by solving for the distribution that generates them, which is both computationally demanding and an unrealistic description of expectations formation. Moll (2025) suggests that agents should instead forecast prices directly, an approach related to Krusell and Smith (1998)'s perceived law of motion (PLM).[3] In *"Non-Parametric Expectations in Heterogeneous-Agent Models,"* with Pavel Sevcik (UQAM), we pursue this idea non-parametrically: agents forecast prices and other observables with Gaussian process regression, which also quantifies how uncertain each forecast is.

The resulting forecasts and aggregate dynamics differ most from those under standard PLMs in two cases: in finite samples, where posterior variance remains wide in rarely visited regions of the state space, and when the aggregate state cannot be recovered from the observables. The direction of the effect is ambiguous: posterior variance above its irreducible level raises precautionary savings, whereas confounding shocks, for instance transitory and persistent ones, can push the other way. Our approach thus lets uncertainty about forecasts shape behavior in HA models, a feature absent under standard PLMs.

[3]The idea is close to how RI models are solved.

Other Research

News and Noise in Expectations

Expectations about future productivity are a natural source of business cycle fluctuations, but agents receive both genuine information about future productivity (news) and signals unrelated to it (noise), and the literature disagrees on how much noise matters. In *"Noisy Expectations and Business Cycles: Identification with Local Projections,"* with Alain Guay (UQAM) and Patrick Fève (TSE), we trace that disagreement to two distinct issues. The first concerns identification: existing procedures exploit different information sets and therefore need not recover the same object. We characterize when a structural impulse response can be recovered from a finite window of future observations, and develop a local-projection framework that recovers, on the same data, the responses to either the Wold innovation of total factor productivity (TFP) growth or the structural news shock. The second issue is economic: how noisy a signal is and how much noise shocks matter for fluctuations are distinct quantities, and we propose long-run statistics for the news content of expectation signals. On U.S. data, roughly 37% of the flow of new information in consumer-confidence signals is noise, yet noise shocks have only small and transitory effects on real activity, while news shocks generate persistent expansions: expectations can be substantially noisy without noise being an important source of business cycle fluctuations.

Attention and Inflation Dynamics

Attention to inflation is not constant: it rises when inflation is high and is shaped by media coverage, which frames the information agents act on. In *"Inflation, Attention and Expectations,"* with Massimiliano Marcellino (Bocconi) and Dalibor Stevanovic (UQAM), we measure attention with Google Trends searches for "inflation" and the share of *Wall Street Journal* headlines about inflation. We show that both measures explain inflation expectations, suggesting that attention affects inflation through expectations. To rationalize this channel, we build a dispersed-information price-setting model in which firms use a public signal only when attention is high, which anchors their higher-order beliefs and mutes the real effects of monetary policy. A threshold VAR in which attention determines the regime confirms that the real effects of monetary policy are smaller when attention is high. Taken together, our findings suggest that monetary policy has less traction on real activity precisely when attention to inflation is high.

Concluding Remarks & Future Research

Taken together, my work models expectations as the outcome of costly information processing and learning, rather than imposing how they are formed, and asks what this implies for the propagation of shocks and for policy. Going forward, I plan to take these frameworks beyond local approximations. The RI-DSGE models above are solved around the non-stochastic steady state, where losses are quadratic and optimal signals Gaussian. Outside this linear-quadratic case, however, optimal signals are often discrete even when the state of the world is continuous (Matějka, 2016). In fully nonlinear RI-DSGE models, discrete signals could explain sharp movements in aggregates that fundamentals alone cannot account for.

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